graph.

[6] Check out chapters 12 and 13.

[7] It is useful to have some flexibility about the region from which points are drawn, because this way we can get better results depending on the size of the graph, avoiding too dense and too sparse embeddings.

[8] Top-left and bottom-right, following the way we index the screen's rows and columns in most programming languages.

[9] If you remember chapter 14, a loop is an edge that starts and ends at the same vertex.

[10] Here we are not talking about moving our imaginary marble on the surface of the cost function: this is a generic description fitting all this category of optimization problems. For the particular problem of the minimum intersection embedding, the way we explore the surrounding are of the cost function is, indeed, moving each vertex around.

[11] Note that there are higher order derivates too, although here we'll stop at the first order derivative.

[12] Not the most famous one! Although one might argue that this particular theorem is even more important.

[13] Tuning algorithm's hyper-parameters, as we'll see, it's one of the challenges of using heuristics. There is no single value for these parameters that works with all problems and instances, and it's usually even hard to automate: it's an area where the experience of the algorithmist.

[14] Nothing more than a function from a domain to a range, really: no matter how complex this function can be, it's still a deterministic mapping between an input (possibly multidimensional) and an output.

[15] Usually we try different models and automate the choice based on how well they perform, but this is far beyond the scope of this discussion.

[16] Actually, a subset of the dataset called training set. This is not the place to go into the details of how and why we choose it, just remember it's quite important to leave some dataset points out for later testing, to assess the quality of the model.

[17] When we use a linear model: in general, we denote with `W` or `θ` the vector of parameters for the model; remember, these parameters are actually the objective of the learning algorithm.

[18] The mathematical solution involves finding the zeroes of the differential equations describing a system.

[19] In reality, electrical forces between particles are order of magnitude stronger than gravity, of course. But the purpose of the analogy is not to be exact, just to be lifelike, intuitive, and reuse a well-studied framework.

[20] Square roots are notoriously expensive operations to perform.